SUMANTH R

Chennai, Tamil Nadu
B.Tech. in Computer Science and Engineering (AIML)
SRM Institute of Science and Technology

+91-9344249005
rsumanth2005@gmail.com
LinkedIn
GitHub

Education

Degree	Institute	CGPA	Duration
B.Tech., Computer Science and Engineering (AIML)	SRM Institute of Science and Technology, Chennai, Tamil Nadu	8.98 / 10	*2023 – 2027*

Experience

- **Works Applications India Pvt. Ltd.** *Dec 2025 – Jan 2026*
  Software Engineering Intern On-site
  - Built and deployed a full-stack application using **Java, Spring Boot, MySQL, and React**, developing 5+ REST APIs for business workflows.
  - Designed database workflows and optimized **SQL queries and indexing**, reducing backend response time by **25%** during testing.
  - Implemented backend-to-frontend integrations and **API-driven workflows**, contributing across application logic, data, and user-facing features.

- **All India Council for Technical Education (AICTE)** *July 2024 – September 2024*
  Google AI/ML Virtual Intern Remote
  - Developed and evaluated **8+ machine learning models** across supervised and unsupervised learning tasks using 5+ datasets.
  - Applied **data preprocessing, feature engineering, EDA, and hyperparameter tuning** to improve model performance.
  - Analyzed real-world AI use cases and applied **bias mitigation** techniques to improve model reliability and evaluation.

Projects

- **Multi-Tenant Education Lending Platform** *2026*
  Technologies: Python, FastAPI, React.js, TypeScript, PostgreSQL, MongoDB, AWS, Docker
  - Built a configurable lending platform with **multi-tenant architecture**, role-based access, loan workflows, and REST APIs for borrower and operations teams.
  - Designed relational data models and optimized **SQL queries, indexing, and database access** for reliable transaction and application workflows.
  - Developed React/TypeScript interfaces and containerized FastAPI services with **Docker**, deploying application components on AWS.

- **Distributed Job Scheduler** *2026*
  Technologies: Python, FastAPI, PostgreSQL, Redis, Docker, AWS
  - Developed a distributed job scheduling system with **REST APIs, task queues, worker coordination, retries, and failure handling**.
  - Implemented persistent job tracking, scheduling, concurrency control, and **idempotent execution** using PostgreSQL and Redis.
  - Containerized services with Docker and designed the system for **horizontal scalability, fault tolerance, and reliable job execution**.

- **AI-Powered SQL Learning Platform** *2025*
  Technologies: React.js, TypeScript, Node.js, SQL, MongoDB, REST APIs
  - Built an interactive full-stack platform for practicing SQL through **progressive query challenges, user profiles, and leaderboard workflows**.
  - Developed REST APIs and database workflows for **query validation, progress tracking, and persistent user data**.
  - Integrated AI-assisted hints to provide contextual guidance while keeping **SQL correctness and application logic** deterministic.

Publications

- **Deep Learning Based Brain Tumor Classification using Orange Data Analytics** *2026*
  IEEE International Conference (SCEECS 2026) View Paper
  - Developed and evaluated a deep learning pipeline using a **16,000+ MRI image dataset** aggregated from Kaggle, BRISC, and FigShare.
  - Applied **data preprocessing, PCA, augmentation, and stratified 10-fold cross-validation**, reducing feature dimensionality by **75%** while retaining **95% variance**.
  - Achieved **96%+ accuracy** and **0.997 AUC**, improving performance by **7% over baseline methods** and MCC by **8%**.

Achievements

- Selected as a **Software Engineering Fellow at Headstarter AI**, working on real-world software engineering and AI projects.
- Secured **3rd Place** in the **CodeMavens 2025 Hackathon** organized by the ACM-W SRM Student Chapter.
- Published research on **deep learning-based brain tumor classification** at the **IEEE International Conference (SCEECS 2026)**.

Technical Skills

- **Languages**: Python, Java, C++, JavaScript, TypeScript, SQL
- **Backend Engineering**: FastAPI, Node.js, Spring Boot, REST APIs, API Design, Authentication, Authorization, Microservices
- **Frontend Engineering**: React.js, TypeScript, HTML, CSS, Tailwind CSS, Responsive UI
- **Databases**: PostgreSQL, MySQL, MongoDB, SQL, NoSQL, Data Modeling, Query Optimization, Indexing
- **Cloud / Infrastructure**: AWS, Docker, Kubernetes, Linux, CI/CD, Git, GitHub
- **Systems / Engineering**: Data Structures & Algorithms, OOP, DBMS, System Design, Distributed Systems, Scalability, Concurrency, Caching
- **AI Engineering**: Generative AI, LLMs, RAG, LangChain, Prompt Engineering, AI-Assisted Development